

COURSE PLAN

PYTHON PROGRAMMING LAB

Prepared by

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UNIT-WISE COURSE PLAN

PYTHON PROGRAMMING LAB

Total Hours: 45 (15 sessions × 3 hours)

Software / Tools: Python 3, Jupyter Notebook / IDLE, NumPy, Pandas, SciPy, Matplotlib, scikit-learn (Anaconda distribution); Linux or Windows

Reference Books: (1) Wesley J. Chun, *Core Python Programming*, 2/e, Prentice Hall; (2) Chris Albon, *Machine Learning with Python Cookbook*, O'Reilly, 2018; (3) Mark Summerfield, *Programming in Python 3*, 2/e, Addison-Wesley; (4) Phuong Vo.T.H, Martin Czygan, *Getting Started with Python Data Analysis*, Packt; (5) Armando Fandango, *Python Data Analysis*, Packt; (6) Magnus Vilhelm Persson, Luiz Felipe Martins, *Mastering Python Data Analysis*, Packt; (7) Sebastian Raschka, Vahid Mirjalili, *Python Machine Learning*, Packt

UNIT I: Python Data Structures, Text and File Handling — 12 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Lab introduction; Python programs on Lists and Dictionaries – creation, indexing, slicing, methods, comprehensions	3	Lecture / Demonstration	Word-frequency counter; student-marks dictionary
2	Python programs on Searching and Sorting – linear and binary search; bubble, selection, insertion, merge and quick sort	3	Demonstration / Guided Practice	Implement and compare techniques on lists
3	Python programs on Text Handling – string methods, formatting, regular expressions	3	Worked Examples / Hands-on Practice	Count words, vowels and palindromes; validate e-mail IDs
4	Python programs on File Handling – reading, writing, appending text and CSV files; exception handling	3	Guided Hands-on Practice	Copy, merge and search files; process a CSV file

UNIT II: Statistical Measures — 6 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Python programs for calculating Mean, Mode, Median, Variance and Standard Deviation	3	Worked Examples / Hands-on Practice	Compute measures manually and verify with the statistics module
2	Python programs for Karl Pearson Coefficient of Correlation and Rank Correlation	3	Worked Examples / Hands-on Practice	Compute correlations and verify with NumPy / SciPy

UNIT III: NumPy and Pandas — 6 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Python programs on NumPy Arrays – creation, indexing, slicing, broadcasting; Linear algebra with NumPy	3	Demonstration / Guided Practice	Matrix operations, determinant, inverse, eigenvalues, solving linear equations
2	Python programs for creation and manipulation of Data Frames using Pandas – selection, filtering, grouping, merging	3	Demonstration / Hands-on Practice	Load a CSV dataset and summarise it with Pandas

UNIT IV: Data Visualization with Matplotlib — 6 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Simple Line Plots; Adjusting the Plot – line colours and styles, axes limits, labelling; Simple Scatter Plots; Histograms; Boxplot	3	Demonstration / Guided Practice	Plot a dataset in different chart types
2	Customizing Plot Legends; Choosing Elements for the Legend; Multiple Legends; Customizing Colorbars; Multiple Subplots; Text and Annotation; Customizing Ticks	3	Worked Examples / Hands-on Practice	Build a multi-panel annotated figure

UNIT V: Machine Learning with Python — 12 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Data pre-processing – handling missing values, handling categorical data, bringing features to the same scale, selecting meaningful features	3	Demonstration / Guided Practice	Clean and prepare a real dataset with Pandas and scikit-learn
2	Compressing data via dimensionality reduction – Principal Component Analysis (PCA)	3	Worked Examples / Hands-on Practice	Apply PCA to the Wine dataset; plot explained variance
3	Data Clustering – K-Means, hierarchical clustering; elbow method	3	Demonstration / Hands-on Practice	Cluster the Iris dataset and visualise the clusters
4	Classification – k-NN, Decision Tree, Logistic Regression; Model Evaluation using K-fold cross-validation	3	Case Study / Hands-on Practice	Train classifiers and compare K-fold cross-validation accuracy

UNIT VI: Revision and Assessment — 3 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Revision, Record Evaluation and Internal Lab Examination	3	Viva-voce / Practical Test	Program execution, record submission and viva

Overall Hour Distribution

Unit	Unit Title	Hours	Percentage
I	Python Data Structures, Text and File Handling	12	26.67%
II	Statistical Measures	6	13.33%
III	NumPy and Pandas	6	13.33%
IV	Data Visualization with Matplotlib	6	13.33%
V	Machine Learning with Python	12	26.67%
VI	Revision and Assessment	3	6.67%
	GRAND TOTAL	45	100%

General Modes of Teaching

- Each laboratory session is of 3 hours: a short briefing/demonstration by the faculty, followed by individual hands-on implementation by students.
- Students prepare the algorithm/design before the session, and execute, test and debug the programs during the session.
- Programs are developed in Jupyter Notebook so that code, output and plots are recorded together.
- Standard datasets (Iris, Wine and CSV files) are used so that statistical, visualization and machine-learning results can be compared.
- Guided problem solving for complex experiments; additional exercises are given to fast learners.
- Continuous assessment through observation book, lab record, program output and viva-voce in every session.
- The final session is for revision, record evaluation and the internal laboratory examination.